A Mizar article in the style of Formalized Mathematics

Unary Primitive Recursive Functions

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Summary. A basis of unary functions, given by initial functions and operators, is suitable when its closure is the set of unary primitive recursive functions. R. M. Robinson proved that $\langle \mathrm{S}, \mathrm{E}, F+G, FG, F^{\square}\rangle$ is suitable [7], where $\mathrm{E}(x) = x - \lfloor\sqrt{x}\rfloor^2$ and $F^{\square}$ is pure iteration. We formalize Section 6 of [9], which replaces the binary operator $F+G$ by the unary operator $F^{+}(x) = F(x) + x$: the bases $\langle \overline{1}, X, F^{+}, FG, F^{\square}\rangle$, for X the excess E or one of the inverses K, L of Cantor's pairing function, and $\langle X, F^{+}, FG, F^{\square(1)}\rangle$, with iteration started at 1, generate the same closure as Robinson's basis. We also answer the question left open in the paper for iteration started at any $a \geq 2$, show that a shifted operator $F^{+b}(x) = F(x) + x + b$ can replace both F^{+} and $\overline{1}$ exactly when $b \geq 1$, and prove the corresponding results for the unary general recursive functions, with iteration replaced by inversion as in the basis of J. Robinson [6].

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INTRODUCTION

A class of unary functions $\mathbb{N} \to \mathbb{N}$ can be described by a basis $\langle F_1, \ldots, F_n, \text{operators}\rangle$: its closure is the least set of functions that contains $F_1, \ldots, F_n$ and is closed under the operators. R. M. Robinson [7] showed that the unary primitive recursive functions are the closure of $\langle \mathrm{S}, \mathrm{E}, F+G, FG, F^{\square}\rangle$, where S is the successor, $\mathrm{E}(x) = x - \lfloor\sqrt{x}\rfloor^2$ is the excess over a square, FG is composition and $F^{\square}$ is pure iteration: $F^{\square}(0) = 0$ and $F^{\square}(x+1) = F(F^{\square}(x))$. Section 6 of [9] removes the binary operator $F+G$: the basis $\langle \overline{1}, \mathrm{E}, F^{+}, FG, F^{\square}\rangle$, with $F^{+}(x) = F(x) + x$, is suitable (Theorem 6.7); so are the bases with the inverses K and L of Cantor's pairing function in place of E (Theorem 6.8, after R. M. Robinson [8]); and, with iteration started at 1, the constant $\overline{1}$ can be dropped (Remark 6.9). This article formalizes that section in the Mizar system [2, 5]. The proofs follow an earlier formalization of the same paper in Isabelle/HOL.

The notion of a primitive recursive unary function is not available in the Mizar Mathematical Library for functions in $\mathbb{N}^{\mathbb{N}}$ [4], so the results are stated as equalities of closures. A basis is given by a set of initial functions, a set of unary operations and a set of binary operations on $\mathbb{N}^{\mathbb{N}}$ [3], and $\mathrm{clos}(I, U, B)$ is the least subset of $\mathbb{N}^{\mathbb{N}}$ that contains I and is closed under U and B (Section 4). Theorem 6.7 becomes $\mathrm{C}_6 = \mathrm{RobinsonClos}$ (82), where C_6 and RobinsonClos are the closures of the two bases above; the proof of the paper, through the basis of its Lemma 3.3, is (83). Lemmas 6.2–6.6, which build F^{-}, $F \oplus G$, $F \ominus G$, $F \otimes G$, $F + G$ and a difference out of F^{+}, are (49)–(75). Theorem 6.8 is (129) and Remark 6.9 is (157).

The remaining sections answer the questions that the paper leaves open. For every $a \geq 1$, the bases $\langle X, F^{+}, FG, F^{\square(a)}\rangle$ with iteration started at a generate C_6, for $X \in \{\mathrm{E}, \mathrm{K}, \mathrm{L}\}$ (198). The basis $\langle \mathrm{E}, F^{+b}, FG, F^{\square}\rangle$ with the shifted operator $F^{+b}(x) = F(x) + x + b$ generates C_6 if and only if $b \geq 1$ (207). J. Robinson [6] showed that the unary general recursive functions are the closure ClosJR of $\langle \mathrm{S}, \mathrm{E}, F+G, FG, F^{-1}\rangle$, where $F^{-1}(x)$ is the least y with $F(y) = x$ for F onto (we take $F^{-1} = F$ otherwise). The bases $\langle \overline{1}, X, F^{+}, FG, F^{-1}\rangle$ generate ClosJR (289), the constant cannot be dropped (297), the successor can replace it (304), the shifted basis $\langle X, F^{+b}, FG, F^{-1}\rangle$ generates ClosJR if and only if $b \geq 1$ (343), and so does the basis with a constant $\overline{n}$ in place of $\overline{1}$, for iteration (357) and for inversion (358).

Throughout, the functions of the paper are written in roman type: I is the identity, $\overline{n}$ the constant function with value n, $\mathrm{D}(x) = 2x$, $\mathrm{M}_r(x) = rx$, $\mathrm{Sq}(x) = x^2$, $\mathrm{Hf}(x) = \lfloor x/2\rfloor$, $\mathrm{Rt}(x) = \lfloor\sqrt{x}\rfloor$, O the cosignum, Sgn the signum, Q the characteristic function of the squares, and $\mathrm{A}(x) = x(x+1)/2$, V, K, L the functions of Cantor's pairing. FG is the composition $x \mapsto F(G(x))$, $\dot{-}$ is the arithmetic difference [1], $F^{\square(a)}$ is iteration started at a, and an operator applied to a function is written as in the paper (F^{+}, $F^{\square}$, F^{-1}, $F \oplus G$, …); standing alone, it is written $\cdot^{+}$, $\cdot^{\square}$, $\circ$, $+$, … The functions named expr… are the expressions of the paper, such as $F^{-} \equiv \mathrm{E}\,\mathrm{S}\,(\mathrm{M}_3 F \mathrm{E})^{+}(\mathrm{Sq}\,F^{+})^{+}$ of Lemma 6.2, and the sets named Clos… are closures of bases.

1. PRELIMINARIES

From now on x, y, n, m, k, r, s denote natural numbers.

From now on F, G, H denote functions from $\mathbb{N}$ into $\mathbb{N}$.

The scheme *UFunEx* deals with a unary functor $\mathcal{T}$ yielding a natural number, and states that there exists a function f from $\mathbb{N}$ into $\mathbb{N}$ such that for every natural number x, $f(x) = \mathcal{T}(x)$.

The scheme *UFunUq* deals with a unary functor $\mathcal{T}$ yielding a natural number and functions $\mathcal{F}$, $\mathcal{G}$ from $\mathbb{N}$ into $\mathbb{N}$, and states that $\mathcal{F} = \mathcal{G}$ provided for every natural number x, $\mathcal{F}(x) = \mathcal{T}(x)$, and for every natural number x, $\mathcal{G}(x) = \mathcal{T}(x)$.

Let F be a function from $\mathbb{N}$ into $\mathbb{N}$, and x be a natural number. One can check that $F(x)$ is natural. Now we state the proposition:

(1) If (for every natural number x holds $F(x) = G(x)$), then $F = G$.

Let G, F be functions from $\mathbb{N}$ into $\mathbb{N}$. One can check that $F\,G$ is a function from $\mathbb{N}$ into $\mathbb{N}$.

The scheme *UOpEx* deals with a binary functor $\mathcal{T}$ from the functions from $\mathbb{N}$ into $\mathbb{N}$ and the natural numbers into the natural numbers, and states that there exists a unary operation u on $\mathbb{N}^{\mathbb{N}}$ such that for every function F from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x, $u(F)(x) = \mathcal{T}(F, x)$.

The scheme *UOpUq* deals with a binary functor $\mathcal{T}$ as in *UOpEx* and unary operations $\mathcal{U}$, $\mathcal{V}$ on $\mathbb{N}^{\mathbb{N}}$, and states that $\mathcal{U} = \mathcal{V}$ provided for every function F from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x, $\mathcal{U}(F)(x) = \mathcal{T}(F, x)$ and $\mathcal{V}(F)(x) = \mathcal{T}(F, x)$.

The scheme *BOpEx* deals with a ternary functor $\mathcal{T}$ from two functions from $\mathbb{N}$ into $\mathbb{N}$ and a natural number into the natural numbers, and states that there exists a binary operation b on $\mathbb{N}^{\mathbb{N}}$ such that for all functions F, G from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x, $b(F, G)(x) = \mathcal{T}(F, G, x)$.

The scheme *BOpUq* deals with a ternary functor $\mathcal{T}$ as in *BOpEx* and binary operations $\mathcal{B}$, $\mathcal{C}$ on $\mathbb{N}^{\mathbb{N}}$, and states that $\mathcal{B} = \mathcal{C}$ provided for all functions F, G from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x, $\mathcal{B}(F, G)(x) = \mathcal{T}(F, G, x)$ and $\mathcal{C}(F, G)(x) = \mathcal{T}(F, G, x)$.

2. BASIC AND ARITHMETIC FUNCTIONS

The functor I yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 1) for every natural number x holds $it(x) = x$.

Let n be a natural number. The functor $\overline{n}$ yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 2) for every natural number x holds $it(x) = n$.

The functor S yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 3) for every natural number x holds $it(x) = x + 1$.

The functor D yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 4) for every natural number x holds $it(x) = 2 \cdot x$.

The functor Sq yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 5) for every natural number x holds $it(x) = x \cdot x$.

The functor Hf yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 6) for every natural number x holds $it(x) = x \operatorname{div} 2$.

Now we state the propositions:

(2) Let us consider a natural number x. Then there exists a natural number r such that $r \cdot r \leq x$ and $x < (r+1) \cdot (r+1)$.

(3) If $r \cdot r \leq x$ and $x < (r+1) \cdot (r+1)$ and $s \cdot s \leq x$ and $x < (s+1) \cdot (s+1)$, then $r = s$.

The functor Rt yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 7) for every natural number x holds $it(x) \cdot it(x) \leq x$ and $x < (it(x)+1) \cdot (it(x)+1)$.

Now we state the proposition:

(4) If $r \cdot r \leq x$ and $x < (r+1) \cdot (r+1)$, then $\mathrm{Rt}(x) = r$. The theorem is a consequence of (3).

The functor E yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 8) for every natural number x holds $it(x) = x \dot{-} \mathrm{Rt}(x) \cdot \mathrm{Rt}(x)$.

The functor O yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 9) for every natural number x holds (if $x = 0$, then $it(x) = 1$) and (if $x \neq 0$, then $it(x) = 0$).

The functor Sgn yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 10) for every natural number x holds (if $x = 0$, then $it(x) = 0$) and (if $x \neq 0$, then $it(x) = 1$).

3. Operators

The functor $\cdot^{+}$ yielding a unary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 11) for every function F from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x holds $it(F)(x) = F(x) + x$.

The functor $+$ yielding a binary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 12) for every functions F, G from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x holds $it(F,G)(x) = F(x) + G(x)$.

The functor $\dot{-}$ yielding a binary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 13) for every functions F, G from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x holds $it(F,G)(x) = F(x) \dot{-} G(x)$.

The functor $\circ$ yielding a binary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 14) for every functions F, G from $\mathbb{N}$ into $\mathbb{N}$ holds $it(F,G) = F\,G$.

The functor $\cdot^{\square}$ yielding a unary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 15) for every function F from $\mathbb{N}$ into $\mathbb{N}$ holds $it(F)(0) = 0$ and for every natural number x holds $it(F)(x+1) = F(it(F)(x))$.

4. BASES AND CLOSURES

Let I, U, B be sets, and X be a subset of $\mathbb{N}^{\mathbb{N}}$. We say that X is closed under I, U, B if and only if

(Def. 16) (i) $I \subseteq X$, and

(ii) for every unary operation u on $\mathbb{N}^{\mathbb{N}}$ and every function F from $\mathbb{N}$ into $\mathbb{N}$ such that $u \in U$ and $F \in X$ holds $u(F) \in X$, and

(iii) for every binary operation b on $\mathbb{N}^{\mathbb{N}}$ and every functions F, G from $\mathbb{N}$ into $\mathbb{N}$ such that $b \in B$ and $F \in X$ and $G \in X$ holds $b(F, G) \in X$.

Let I, U, B be sets. The functor $\mathrm{clos}(I, U, B)$ yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 17) for every function F from $\mathbb{N}$ into $\mathbb{N}$ holds $F \in it$ iff for every subset X of $\mathbb{N}^{\mathbb{N}}$ such that X is closed under I, U, B holds $F \in X$.

From now on I, U, B denote sets. Now we state the propositions:

(5) If $F \in I$, then $F \in \mathrm{clos}(I, U, B)$.

(6) Let us consider a unary operation u on $\mathbb{N}^{\mathbb{N}}$. Suppose $u \in U$ and $F \in \mathrm{clos}(I, U, B)$. Then $u(F) \in \mathrm{clos}(I, U, B)$.

(7) Let us consider a binary operation b on $\mathbb{N}^{\mathbb{N}}$. Suppose $b \in B$ and $F \in \mathrm{clos}(I, U, B)$ and $G \in \mathrm{clos}(I, U, B)$. Then $b(F, G) \in \mathrm{clos}(I, U, B)$.

(8) Let us consider a subset X of $\mathbb{N}^{\mathbb{N}}$. Suppose X is closed under I, U, B. Then $\mathrm{clos}(I, U, B) \subseteq X$.

(9) If $I \subseteq \mathbb{N}^{\mathbb{N}}$, then $\mathrm{clos}(I, U, B)$ is closed under I, U, B. The theorem is a consequence of (5), (6), and (7).

5. ARITHMETIC FACTS ABOUT Rt AND E

Now we state the propositions:

(10) If $k \leq 2 \cdot m$, then $\mathrm{Rt}(m \cdot m + k) = m$. The theorem is a consequence of (4).

(11) $\mathrm{Rt}(m \cdot m) = m$. The theorem is a consequence of (4).

(12) $\mathrm{E}(x) + \mathrm{Rt}(x) \cdot \mathrm{Rt}(x) = x$.

(13) If $k \leq 2 \cdot m$, then $\mathrm{E}(m \cdot m + k) = k$. The theorem is a consequence of (10).

(14) $\mathrm{E}(x) \leq x$. The theorem is a consequence of (12).

(15) Let us consider natural numbers a, b. Suppose $b \leq a$. Then $\mathrm{E}((a + b) \cdot (a + b) + 3 \cdot a + b + 1) = a \dot{-} b$. The theorem is a consequence of (13).

6. THE BASIS $\langle \overline{1}, \mathrm{E}, F^+, FG, F^\square \rangle$

From now on q, t denote natural numbers. Now we state the proposition:

(16) $(F\,G)(x) = F(G(x))$.

The functor C_6 yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 18) $\mathrm{clos}(\{\overline{1}, \mathrm{E}\}, \{\cdot^+, \cdot^\square\}, \{\circ\})$.

Now we state the propositions:

(17) $\overline{1} \in \mathrm{C}_6$. The theorem is a consequence of (5).

(18) $\mathrm{E} \in \mathrm{C}_6$. The theorem is a consequence of (5).

(19) If $F \in \mathrm{C}_6$, then $F^+ \in \mathrm{C}_6$. The theorem is a consequence of (6).

(20) If $F \in \mathrm{C}_6$, then $F^\square \in \mathrm{C}_6$. The theorem is a consequence of (6).

(21) If $F \in \mathrm{C}_6$ and $G \in \mathrm{C}_6$, then $F\,G \in \mathrm{C}_6$. The theorem is a consequence of (7).

(22) (i) $\mathrm{E}(0) = 0$, and
(ii) $\mathrm{E}(1) = 0$, and
(iii) $\mathrm{E}(2) = 1$, and
(iv) $\mathrm{E}(3) = 2$, and
(v) $\mathrm{E}(4) = 0$.

(23) $\mathrm{E}(x) = 0$ if and only if there exists k such that $x = k \cdot k$.

The functor Mod_3 yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 19) for every natural number x holds $it(x) = x \bmod 3$.

The functor M_3 yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 20) for every natural number x holds $it(x) = 3 \cdot x$.

The functor R yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 21) for every natural number x holds $it(x) = x + 2 \cdot \mathrm{Rt}(x)$.

The functor Q yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 22) for every natural number x holds (if (there exists a natural number k such that $x = k \cdot k$), then $it(x) = 1$) and (if (there exists no natural number k such that $x = k \cdot k$), then $it(x) = 0$).

Now we state the propositions:

(24) $\overline{1}^+ = \mathrm{S}$. The theorem is a consequence of (1).

(25) $\overline{1}^\square = \mathrm{Sgn}$.

(26) $\mathrm{E}\,\overline{1} = \overline{0}$. The theorem is a consequence of (1).

(27) $(\overline{0}^+)^+ = \mathrm{D}$. The theorem is a consequence of (1).

(28) $\mathrm{D}^+ = \mathrm{M}_3$. The theorem is a consequence of (1).

(29) $\mathrm{E\,D\,S\,Sgn} = \mathrm{O}$.

(30) $\mathrm{O\,E} = \mathrm{Q}$.

(31) $(\mathrm{E\,S\,S})(x) = \mathrm{E}(x + 2)$.

(32) $(\mathrm{E\,S\,S})^\square = \mathrm{Mod}_3$.

(33) If $m \cdot m < x$ and $x < (m+1) \cdot (m+1)$, then there exists no k such that $x = k \cdot k$.

(34) $(\mathrm{D}\,\mathrm{Q}\,\mathrm{S}\,\mathrm{S}\,\mathrm{S})^{+}(\mathrm{S}(\mathrm{R}(n))) = \mathrm{R}(n+1)$.

(35) $((\mathrm{D}\,\mathrm{Q}\,\mathrm{S}\,\mathrm{S}\,\mathrm{S})^{+}\,\mathrm{S})^{\square} = \mathrm{R}$.

(36) $(\mathrm{S}\,\mathrm{R})^{\square} = \mathrm{Sq}$.

(37) $\mathrm{S} \in \mathrm{C}_6$. The theorem is a consequence of (17), (19), and (24).

(38) $\mathrm{Sgn} \in \mathrm{C}_6$. The theorem is a consequence of (17), (20), and (25).

(39) $\overline{0} \in \mathrm{C}_6$. The theorem is a consequence of (17), (18), (21), and (26).

(40) $\mathrm{D} \in \mathrm{C}_6$. The theorem is a consequence of (39), (19), and (27).

(41) $\mathrm{M}_3 \in \mathrm{C}_6$. The theorem is a consequence of (40), (19), and (28).

(42) $\mathrm{O} \in \mathrm{C}_6$. The theorem is a consequence of (18), (40), (21), (37), (38), and (29).

(43) $\mathrm{Q} \in \mathrm{C}_6$. The theorem is a consequence of (42), (18), (21), and (30).

(44) $\mathrm{Mod}_3 \in \mathrm{C}_6$. The theorem is a consequence of (18), (37), (21), (20), and (32).

(45) $\mathrm{R} \in \mathrm{C}_6$. The theorem is a consequence of (40), (43), (21), (37), (19), (20), and (35).

(46) $\mathrm{Sq} \in \mathrm{C}_6$. The theorem is a consequence of (37), (45), (21), (20), and (36).

7. Definition 6.1 and Lemmas 6.2–6.6

The functor $\cdot^{-}$ yielding a unary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 23) for every function F from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x holds $it(F)(x) = F(x) \dot{-} x$.

The functor $\oplus$ yielding a binary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 24) for every functions F, G from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x holds $it(F,G)(x) = F(x) + G(x) \cdot G(x)$.

The functor $\ominus$ yielding a binary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 25) for every functions F, G from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x holds $it(F,G)(x) = F(x) \dot{-} G(x) \cdot G(x)$.

The functor $\otimes$ yielding a binary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 26) for every functions F, G from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x holds $it(F,G)(x) = F(x) \cdot G(x)$.

Let F be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor exprMinusI(F) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 27) $\mathrm{E}\,\mathrm{S}\,(\mathrm{M}_3\,F\,\mathrm{E})^{+}\,(\mathrm{Sq}\,F^{+})^{+}$.

Now we state the propositions:

(47) If $F \in \mathrm{C}_6$, then exprMinusI(F) $\in \mathrm{C}_6$. The theorem is a consequence of (18), (37), (21), (41), (19), and (46).

(48) If $x \leq F(x)$, then exprMinusI(F)$(x) = F(x) \dot{-} x$.

(49) If $F \in \mathrm{C}_6$ and (for every x holds $x \leq F(x)$), then $F^- \in \mathrm{C}_6$. The theorem is a consequence of (1) and (47).

(50) $(\mathrm{S}\,\mathrm{Mod}_3^+)^{\square}(2 \cdot q) = 3 \cdot q$ and $(\mathrm{S}\,\mathrm{Mod}_3^+)^{\square}(2 \cdot q + 1) = 3 \cdot q + 1$.

(51) $(\mathrm{S}\,\mathrm{Mod}_3^+)^{\square}(x) = x + (x \operatorname{div} 2)$.

The functor exprHf yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 28) exprMinusI$((\mathrm{S}\,\mathrm{Mod}_3^+)^{\square})$.

The functor exprRt yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 29) Hf exprMinusI(R).

Now we state the propositions:

(52) exprHf = Hf.

(53) exprRt = Rt.

(54) Hf $\in \mathrm{C}_6$ and Rt $\in \mathrm{C}_6$. The theorem is a consequence of (44), (19), (37), (21), (20), (47), (52), (45), and (53).

(55) Hf $\in \mathrm{C}_6$. The theorem is a consequence of (54).

(56) Rt $\in \mathrm{C}_6$. The theorem is a consequence of (54).

(57) $n \leq n \cdot n$.

Let F, G be functions from $\mathbb{N}$ into $\mathbb{N}$. The functor exprOplus(F, G) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 30) $(\mathrm{exprMinusI}(F)\,\mathrm{E})^+\,(\mathrm{Sq}\,G)^+$.

Let F, G be functions from $\mathbb{N}$ into $\mathbb{N}$. The functor exprOminus(F, G) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 31) $\mathrm{exprMinusI}(F^+\,\mathrm{E})\,(\mathrm{Sq}\,G)^+$.

Now we state the propositions:

(58) If $F \in \mathrm{C}_6$ and $G \in \mathrm{C}_6$, then exprOplus$(F, G) \in \mathrm{C}_6$. The theorem is a consequence of (47), (18), (21), (19), and (46).

(59) If $F \in \mathrm{C}_6$ and $G \in \mathrm{C}_6$, then exprOminus$(F, G) \in \mathrm{C}_6$. The theorem is a consequence of (19), (18), (21), (47), and (46).

(60) If $x \leq G(x)$, then $(\mathrm{Sq}\,G)^+(x) = G(x) \cdot G(x) + x$ and $\mathrm{E}(G(x) \cdot G(x) + x) = x$. The theorem is a consequence of (13).

(61) If $x \leq F(x)$ and $x \leq G(x)$, then exprOplus$(F, G)(x) = F(x) + G(x) \cdot G(x)$. The theorem is a consequence of (60).

(62) If $x \leq G(x)$ and $G(x) \cdot G(x) \leq F(x)$, then exprOminus$(F, G)(x) = F(x) \mathbin{\dot{-}} G(x) \cdot G(x)$.

(63) If $F \in \mathrm{C}_6$ and $G \in \mathrm{C}_6$ and (for every x holds $x \leq F(x)$) and (for every x holds $x \leq G(x)$), then
  (i) $F \oplus G \in \mathrm{C}_6$, and
  (ii) if (for every x holds $G(x) \cdot G(x) \leq F(x)$), then $F \ominus G \in \mathrm{C}_6$.

The functor exprPhat yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 32) exprOminus$(((\mathrm{Sq}^+)^+)^+, \mathrm{S})$.

Now we state the propositions:

(64) exprPhat $\in \mathrm{C}_6$. The theorem is a consequence of (46), (19), (37), and (59).

(65) $\mathrm{exprPhat}(x+1) = x$.

Let F, G be functions from $\mathbb{N}$ into $\mathbb{N}$. The functor exprOtimes(F, G) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 33) Rt exprOminus(Hf Hf exprOminus(Sq exprPhat exprOminus(exprOplus(Sq S Sq $F, G)$), Sq F), Sq G), Sq F).

Now we state the propositions:

(66) If $F \in \mathrm{C}_6$ and $G \in \mathrm{C}_6$, then exprOtimes$(F, G) \in \mathrm{C}_6$.

(67) If $x \leq F(x)$ and $x \leq G(x)$, then exprOtimes$(F, G)(x) = F(x) \cdot G(x)$.

(68) If $F \in \mathrm{C}_6$ and $G \in \mathrm{C}_6$ and (for every x holds $x \leq F(x)$) and (for every x holds $x \leq G(x)$), then $F \otimes G \in \mathrm{C}_6$. The theorem is a consequence of (1) and (66).

Let F, G be functions from $\mathbb{N}$ into $\mathbb{N}$. The functor exprPlus(F, G) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 34) exprMinusI(exprMinusI(Rt exprOplus(exprOplus(D exprOtimes(F^+, G^+), F^+), G^+))).

Let F, G be functions from $\mathbb{N}$ into $\mathbb{N}$. The functor exprMinus(F, G) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 35) E S exprPlus(exprPlus(Sq exprPlus(F, G), M_3 F), G).

Now we state the propositions:

(69) If $F \in \mathrm{C}_6$ and $G \in \mathrm{C}_6$, then exprPlus$(F, G) \in \mathrm{C}_6$.

(70) If $F \in \mathrm{C}_6$ and $G \in \mathrm{C}_6$, then exprMinus$(F, G) \in \mathrm{C}_6$. The theorem is a consequence of (69), (46), (21), (41), (18), and (37).

(71) $\mathrm{exprPlus}(F, G)(x) = F(x) + G(x)$.

(72) $\mathrm{exprPlus}(F, G) = F + G$. The theorem is a consequence of (1).

The functor $\dot{-}_E$ yielding a binary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 36) for every functions F, G from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x holds $it(F, G)(x) = \mathrm{E}((F(x) + G(x)) \cdot (F(x) + G(x)) + 3 \cdot F(x) + G(x) + 1)$.

Now we state the propositions:

(73) If $G(x) \leq F(x)$, then $(F \dot{-}_E G)(x) = F(x) \dot{-} G(x)$. The theorem is a consequence of (15).

(74) $\mathrm{exprMinus}(F, G) = F \dot{-}_E G$.

(75) If $F \in \mathrm{C}_6$ and $G \in \mathrm{C}_6$, then
  (i) $F + G \in \mathrm{C}_6$, and
  (ii) $F \dot{-}_E G \in \mathrm{C}_6$, and
  (iii) if (for every x holds $G(x) \leq F(x)$), then $F \dot{-} G \in \mathrm{C}_6$.

8. THEOREM 6.7

The functor RobinsonClos yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 37) $\mathrm{clos}(\{\mathrm{S}, \mathrm{E}\}, \{\cdot^{\square}\}, \{+, \circ\})$.

Now we state the propositions:

(76) $\mathrm{E}^{\square} = \overline{0}$.

(77) $\mathrm{S}^{\square} = \mathrm{I}$.

(78) $\mathrm{S}\,\overline{0} = \overline{1}$. The theorem is a consequence of (1).

(79) $F^{+} = F + \mathrm{I}$. The theorem is a consequence of (1).

(80) C_6 is closed under $\{\mathrm{S}, \mathrm{E}\}$, $\{\cdot^{\square}\}$, $\{+, \circ\}$.

(81) RobinsonClos is closed under $\{\overline{1}, \mathrm{E}\}$, $\{\cdot^{+}, \cdot^{\square}\}$, $\{\circ\}$.

(82) $\mathrm{C}_6 = \mathrm{RobinsonClos}$. The theorem is a consequence of (80), (8), and (81).

(83) $\mathrm{clos}(\{\mathrm{S}, \mathrm{Sq}, \mathrm{O}, \mathrm{Hf}, \mathrm{Rt}\}, \{\cdot^{\square}\}, \{+, \dot{-}_E, \circ\}) \subseteq \mathrm{C}_6$.

9. CANTOR PAIRING FUNCTIONS

Now we state the proposition:

(84) There exists a natural number a such that $2 \cdot a = x \cdot x + x$.

The functor A yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 38) for every natural number x holds $it(x) = (x \cdot x + x) \operatorname{div} 2$.

Now we state the propositions:

(85) $2 \cdot \mathrm{A}(x) = x \cdot x + x$. The theorem is a consequence of (84).

(86) $\mathrm{A}(x + 1) = \mathrm{A}(x) + x + 1$.

(87) If $m \leq n$, then $\mathrm{A}(m) \leq \mathrm{A}(n)$. The theorem is a consequence of (85).

(88) If $\mathrm{A}(m) < \mathrm{A}(n)$, then $m < n$. The theorem is a consequence of (87).

(89) There exists a natural number t such that $\mathrm{A}(t) \leq x$ and $x < \mathrm{A}(t + 1)$.

(90) If $\mathrm{A}(r) \leq x$ and $x < \mathrm{A}(r + 1)$ and $\mathrm{A}(s) \leq x$ and $x < \mathrm{A}(s + 1)$, then $r = s$.

The functor V yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 39) for every natural number x holds $it(x) = (\mathrm{Rt}(8 \cdot x + 1) \dot{-} 1) \operatorname{div} 2$.

Now we state the propositions:

(91) If $\mathrm{A}(t) \leq x$ and $x < \mathrm{A}(t + 1)$, then $\mathrm{V}(x) = t$.

(92) $\mathrm{A}(\mathrm{V}(x)) \leq x$ and $x < \mathrm{A}(\mathrm{V}(x) + 1)$. The theorem is a consequence of (89) and (91).

The functor K yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 40) for every natural number x holds $it(x) = x \dot{-} \mathrm{A}(\mathrm{V}(x))$.

The functor L yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 41) for every natural number x holds $it(x) = \mathrm{A}(\mathrm{V}(x) + 1) \dot{-} x \dot{-} 1$.

Now we state the propositions:

(93) If $k \leq t$, then

(i) $\mathrm{V}(\mathrm{A}(t)+k)=t$, and
(ii) $\mathrm{K}(\mathrm{A}(t)+k)=k$, and
(iii) $\mathrm{L}(\mathrm{A}(t)+k)=t \dot{-} k$.

(94) $x=\mathrm{A}(\mathrm{V}(x))+\mathrm{K}(x)$ and $\mathrm{K}(x)\leq \mathrm{V}(x)$. The theorem is a consequence of (92) and (86).

(95) $\mathrm{L}(x)=\mathrm{V}(x) \dot{-} \mathrm{K}(x)$. The theorem is a consequence of (94) and (93).

(96) $\mathrm{I}\in \mathrm{C}_6$. The theorem is a consequence of (37), (20), and (77).

(97) $\mathrm{A}\in \mathrm{C}_6$.

(98) $\mathrm{V}\in \mathrm{C}_6$.

(99) $\mathrm{K}\in \mathrm{C}_6$.

(100) $\mathrm{L}\in \mathrm{C}_6$. The theorem is a consequence of (94), (1), (98), (99), and (75).

10. Theorem 6.8

Let G be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor $\mathrm{Clos1}(G)$ yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 42) $\mathrm{clos}(\{\overline{1}, G\}, \{\cdot^{+}, \cdot^{\square}\}, \{\circ\})$.

From now on G_1 denotes a function from $\mathbb{N}$ into $\mathbb{N}$. Now we state the propositions:

(101) $\overline{1}\in \mathrm{Clos1}(G_1)$ and $G_1\in \mathrm{Clos1}(G_1)$. The theorem is a consequence of (5).

(102) If $F\in \mathrm{Clos1}(G_1)$, then $F^{+}\in \mathrm{Clos1}(G_1)$. The theorem is a consequence of (6).

(103) If $F\in \mathrm{Clos1}(G_1)$, then $F^{\square}\in \mathrm{Clos1}(G_1)$. The theorem is a consequence of (6).

(104) If $F\in \mathrm{Clos1}(G_1)$ and $G\in \mathrm{Clos1}(G_1)$, then $F\,G\in \mathrm{Clos1}(G_1)$. The theorem is a consequence of (7).

(105) If $G_1\in \mathrm{Clos1}(H)$, then $\mathrm{Clos1}(G_1)\subseteq \mathrm{Clos1}(H)$.

(106) $\mathrm{S}\in \mathrm{Clos1}(G_1)$. The theorem is a consequence of (101), (102), and (24).

(107) $\mathrm{Sgn}\in \mathrm{Clos1}(G_1)$. The theorem is a consequence of (101), (103), and (25).

The functor $\mathrm{C}_6^{\mathrm{K}}$ yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 43) $\mathrm{Clos1}(\mathrm{K})$.

The functor $\mathrm{C}_6^{\mathrm{L}}$ yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 44) $\mathrm{Clos1}(\mathrm{L})$.

Now we state the proposition:

(108) $\mathrm{C}_6=\mathrm{Clos1}(\mathrm{E})$.

The functor Z yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 45) for every natural number x holds $it(x)=\mathrm{A}(x)+x$.

The functor Y yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 46) for every natural number x holds $it(x)=2\cdot x \dot{-} \mathrm{Rt}(x)$.

Now we state the propositions:

(109) $2 \cdot \mathrm{Z}(x) = x \cdot (x + 3)$.

(110) $\mathrm{A}(1) = 1$. The theorem is a consequence of (85).

(111) $\mathrm{K}\,\overline{1} = \overline{0}$. The theorem is a consequence of (93) and (1).

(112) $(\mathrm{S}\,\mathrm{S}\,\mathrm{K}^+)^\square = \mathrm{Z}$.

(113) Let us consider natural numbers a, b. Suppose $b \leq a$. Then $\mathrm{K}(\mathrm{Z}(a + b) + 2 \cdot a + 3) = a \dot{-} b$.

Let F be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor exprIminusF(F) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 47) $\mathrm{K}\,\mathrm{S}\,\mathrm{S}\,\mathrm{S}\,((\mathrm{Z}\,F^+)^+)^+$.

Let F be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor exprDminusF(F) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 48) $\mathrm{K}\,\mathrm{S}\,\mathrm{S}\,\mathrm{S}\,((((\mathrm{Z}\,(F^+)^+)^+)^+)^+)^+$.

Now we state the propositions:

(114) $(\mathrm{K}\,\mathrm{S}\,\mathrm{S}\,\mathrm{S})(y) = \mathrm{K}(y + 3)$.

(115) If $F(x) \leq x$, then exprIminusF(F)(x) $= x \dot{-} F(x)$.

(116) If $F(x) \leq 2 \cdot x$, then exprDminusF(F)(x) $= 2 \cdot x \dot{-} F(x)$.

(117) $\mathrm{Rt}(x) \leq x$. The theorem is a consequence of (57).

(118) $\mathrm{Y}(n) + \mathrm{Rt}(n) = 2 \cdot n$. The theorem is a consequence of (117).

(119) If $(\mathrm{Rt}(n) + 1) \cdot (\mathrm{Rt}(n) + 1) = n + 1$, then $\mathrm{K}(\mathrm{Y}(n) + 1) = 0$. The theorem is a consequence of (118) and (93).

(120) If $(\mathrm{Rt}(n) + 1) \cdot (\mathrm{Rt}(n) + 1) \neq n + 1$, then $\mathrm{K}(\mathrm{Y}(n) + 1) \neq 0$.

(121) $(\mathrm{Sgn}\,\mathrm{K})^+(\mathrm{S}(\mathrm{Y}(n))) = \mathrm{Y}(n + 1)$.

(122) $((\mathrm{Sgn}\,\mathrm{K})^+\,\mathrm{S})^\square = \mathrm{Y}$.

(123) exprDminusF(Y) = Rt.

(124) $(((\overline{1}^+)^+\,\mathrm{Rt})^+)^\square = \mathrm{Sq}$.

(125) exprIminusF(Sq Rt) = E. The theorem is a consequence of (1).

(126) $\mathrm{E} \in \mathrm{C}_6^{\mathrm{K}}$.

(127) $\mathrm{L}\,\overline{1}^+\,((\overline{1}^+)^+\,\mathrm{L})^+ = \mathrm{K}$.

(128) $\mathrm{K} \in \mathrm{C}_6^{\mathrm{L}}$. The theorem is a consequence of (101), (102), (104), and (127).

(129) $\mathrm{C}_6^{\mathrm{K}} = \mathrm{C}_6$ and $\mathrm{C}_6^{\mathrm{L}} = \mathrm{C}_6$. The theorem is a consequence of (108), (126), (105), (99), (100), and (128).

11. REMARK 6.9: ITERATION STARTED AT 1

The functor $\cdot^{\square(1)}$ yielding a unary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 49) for every function F from $\mathbb{N}$ into $\mathbb{N}$ holds $it(F)(0) = 1$ and for every natural number x holds $it(F)(x+1) = F(it(F)(x))$.

Now we state the propositions:

(130) If (for every x holds $H(x+1) = x$), then $H\,(\mathrm{S}\,F\,H)^{\square(1)} = F^{\square}$.

(131) If $F \in \mathrm{C}_6$ and $G \in \mathrm{C}_6$, then $F \otimes G \in \mathrm{C}_6$.

(132) If $F \in \mathrm{C}_6$, then $F^{\square(1)} \in \mathrm{C}_6$.

Let G be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor ClosB1(G) yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 50) $\mathrm{clos}(\{G\}, \{\cdot^{+}, \cdot^{\square(1)}\}, \{\circ\})$.

Now we state the propositions:

(133) $G_1 \in \mathrm{ClosB1}(G_1)$. The theorem is a consequence of (5).

(134) If $F \in \mathrm{ClosB1}(G_1)$, then $F^{+} \in \mathrm{ClosB1}(G_1)$. The theorem is a consequence of (6).

(135) If $F \in \mathrm{ClosB1}(G_1)$, then $F^{\square(1)} \in \mathrm{ClosB1}(G_1)$. The theorem is a consequence of (6).

(136) If $F \in \mathrm{ClosB1}(G_1)$ and $G \in \mathrm{ClosB1}(G_1)$, then $F\,G \in \mathrm{ClosB1}(G_1)$. The theorem is a consequence of (7).

(137) If $G_1 \in \mathrm{C}_6$, then $\mathrm{ClosB1}(G_1) \subseteq \mathrm{C}_6$.

(138) If $G_1 \in \mathrm{ClosB1}(H)$, then $\mathrm{ClosB1}(G_1) \subseteq \mathrm{ClosB1}(H)$.

(139) If $\overline{1} \in \mathrm{ClosB1}(G_1)$ and $H \in \mathrm{ClosB1}(G_1)$ and (for every x holds $H(x+1) = x$), then $\mathrm{Clos1}(G_1) \subseteq \mathrm{ClosB1}(G_1)$.

(140) If $F(1) = 0$ and $F(0) = 0$, then $F^{\square(1)} = \mathrm{O}$.

(141) If $F(1) = 0$ and $F(0) = 0$, then $F\,\mathrm{O} = \overline{0}$.

(142) $\mathrm{O}\,\overline{0} = \overline{1}$. The theorem is a consequence of (1).

The functor G yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 51) for every natural number x holds $it(x) = (x+1) \cdot (x+1)$.

Now we state the propositions:

(143) $(\mathrm{D}\,\mathrm{Q}\,\mathrm{S}\,\mathrm{S})(y) = 2 \cdot \mathrm{Q}(y+2)$ and $(\mathrm{D}\,\mathrm{Q}\,\mathrm{S}\,\mathrm{S}\,\mathrm{S})(y) = 2 \cdot \mathrm{Q}(y+3)$.

(144) $((\mathrm{D}\,\mathrm{Q}\,\mathrm{S}\,\mathrm{S})^{+}\,\mathrm{S})^{\square(1)}(n) = \mathrm{R}(n) + 1$.

(145) $(((\mathrm{D}\,\mathrm{Q}\,\mathrm{S}\,\mathrm{S})^{+}\,\mathrm{S})^{\square(1)})^{\square(1)} = \mathrm{G}$.

The functor exprPhatI yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 52) $\mathrm{E}\,\mathrm{S}\,\mathrm{S}\,((\mathrm{G}^{+})^{+})^{+}$.

Now we state the propositions:

(146) $\mathrm{exprPhatI}(x+1) = x$.

(147) $\mathrm{ClosB1}(\mathrm{E}) = \mathrm{C}_6$.

The functor H yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 53) for every natural number x holds $it(x) = \mathrm{A}(x+1) + x + 1$.

Now we state the propositions:

(148) $2 \cdot \mathrm{H}(x) = (x+1) \cdot (x+4)$.

(149) $\mathrm{K}(0) = 0$ and $\mathrm{K}(1) = 0$. The theorem is a consequence of (85), (93), and (110).

(150) $(\mathrm{S}\,\mathrm{K}^+\,\mathrm{S})^{\square(1)}(n) = \mathrm{A}(n+1) + n$.

(151) $\mathrm{S}\,(\mathrm{S}\,\mathrm{K}^+\,\mathrm{S})^{\square(1)} = \mathrm{H}$. The theorem is a consequence of (1).

The functor exprPhatII yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 54) $\mathrm{K}\,\mathrm{S}\,\mathrm{S}\,\mathrm{S}\,(\mathrm{H}^+)^+$.

Now we state the propositions:

(152) $\mathrm{exprPhatII}(x+1) = x$.

(153) $\mathrm{ClosB1}(\mathrm{K}) = \mathrm{C}_6$.

(154) $\mathrm{L}(1) = 1$. The theorem is a consequence of (93) and (110).

(155) $\mathrm{L}^{\square(1)} = \overline{1}$.

(156) $\mathrm{ClosB1}(\mathrm{L}) = \mathrm{C}_6$.

(157) (i) $\mathrm{ClosB1}(\mathrm{E}) = \mathrm{C}_6$, and
(ii) $\mathrm{ClosB1}(\mathrm{K}) = \mathrm{C}_6$, and
(iii) $\mathrm{ClosB1}(\mathrm{L}) = \mathrm{C}_6$.

The theorem is a consequence of (147), (153), and (156).

12. Iteration Started at Any $a \geq 1$

From now on a, b, c, d, i, j denote natural numbers.

From now on X, W denote functions from $\mathbb{N}$ into $\mathbb{N}$.

Let a be a natural number. The functor $\cdot^{\square(a)}$ yielding a unary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 55) for every function F from $\mathbb{N}$ into $\mathbb{N}$ holds $it(F)(0) = a$ and for every natural number x holds $it(F)(x+1) = F(it(F)(x))$.

Now we state the proposition:

(158) $\cdot^{\square(1)} = \cdot^{\square(1)}$.

Let r be a natural number. The functor M_r yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by

(Def. 56) for every natural number x holds $it(x) = r \cdot x$.

Now we state the propositions:

(159) (i) $\mathrm{M}_0 = \overline{0}$, and
(ii) $\mathrm{M}_1 = \mathrm{I}$, and
(iii) $\mathrm{M}_2 = \mathrm{D}$, and
(iv) $\mathrm{M}_r^+ = \mathrm{M}_{r+1}$.

(160) If $\overline{0} \in \mathrm{clos}(I, U, B)$ and $\cdot^+ \in U$, then $\mathrm{M}_r \in \mathrm{clos}(I, U, B)$.

(161) If $F \in \mathrm{clos}(I, U, \{\circ\})$ and $G \in \mathrm{clos}(I, U, \{\circ\})$, then $F\,G \in \mathrm{clos}(I, U, \{\circ\})$. The theorem is a consequence of (7).

(162) If $W \in \mathrm{clos}(I, U, \{\circ\})$ and (for every x holds $W(x) = x + c$) and $G \in \mathrm{clos}(I, U, \{\circ\})$, then there exists H such that $H \in \mathrm{clos}(I, U, \{\circ\})$ and for every x holds $H(x) = G(x) + k \cdot c$.

(163) If $W \in \mathrm{clos}(I, U, \{\circ\})$ and (for every x holds $W(x + 1) = x$) and $G \in \mathrm{clos}(I, U, \{\circ\})$, then there exists H such that $H \in \mathrm{clos}(I, U, \{\circ\})$ and for every x, y such that $G(x) = y + k$ holds $H(x) = y$.

(164) If $X \in \mathrm{clos}(I, U, \{\circ\})$ and $X(0) = 0$ and (for every y such that $0 < y$ holds $X(y) < y$) and $G \in \mathrm{clos}(I, U, \{\circ\})$ and (for every x holds $G(x) \leq k$), then there exists H such that $H \in \mathrm{clos}(I, U, \{\circ\})$ and for every x holds $H(x) = 0$.

(165) If (for every y holds $W(y + a) = y$) and (for every x holds $X(x) = x + a$), then $W\,(X\,F\,W)^{\Box(a)} = F^{\Box}$.

(166) If $F \in \mathrm{C}_6$, then $F^{\Box(a)} \in \mathrm{C}_6$.

Let G be a function from $\mathbb{N}$ into $\mathbb{N}$, and a be a natural number. The functor $\mathrm{ClosBa}(G, a)$ yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 57) $\mathrm{clos}(\{G\}, \{\cdot^{+}, \cdot^{\Box(a)}\}, \{\circ\})$.

Now we state the propositions:

(167) $G_1 \in \mathrm{ClosBa}(G_1, a)$. The theorem is a consequence of (5).

(168) If $F \in \mathrm{ClosBa}(G_1, a)$, then $F^{+} \in \mathrm{ClosBa}(G_1, a)$. The theorem is a consequence of (6).

(169) If $F \in \mathrm{ClosBa}(G_1, a)$, then $F^{\Box(a)} \in \mathrm{ClosBa}(G_1, a)$. The theorem is a consequence of (6).

(170) If $F \in \mathrm{ClosBa}(G_1, a)$ and $G \in \mathrm{ClosBa}(G_1, a)$, then $F\,G \in \mathrm{ClosBa}(G_1, a)$. The theorem is a consequence of (161).

(171) $\mathrm{ClosBa}(G_1, 1) = \mathrm{ClosB1}(G_1)$. The theorem is a consequence of (158).

(172) If $G_1 \in \mathrm{C}_6$, then $\mathrm{ClosBa}(G_1, a) \subseteq \mathrm{C}_6$.

(173) If $G_1 \in \mathrm{ClosBa}(H, a)$, then $\mathrm{ClosBa}(G_1, a) \subseteq \mathrm{ClosBa}(H, a)$.

(174) If $\overline{1} \in \mathrm{ClosBa}(X, a)$ and (for every F such that $F \in \mathrm{ClosBa}(X, a)$ holds $F^{\Box} \in \mathrm{ClosBa}(X, a)$), then $\mathrm{Clos1}(X) \subseteq \mathrm{ClosBa}(X, a)$.

(175) If $\overline{0} \in \mathrm{ClosBa}(X, a)$ and $X(0) = 0$ and (for every i such that $1 \leq i$ holds there exists r such that $X(r \cdot (i + 1)) = i$), then there exists W such that $W \in \mathrm{ClosBa}(X, a)$ and $W(k + 1) = 1$ and $W(0) = 0$.

(176) If $2 \leq a$ and $\overline{0} \in \mathrm{ClosBa}(X, a)$ and $X(0) = 0$ and (for every i such that $1 \leq i$ holds there exists r such that $X(r \cdot (i + 1)) = i$), then
  (i) $\mathrm{I} \in \mathrm{ClosBa}(X, a)$, and
  (ii) $\mathrm{D} \in \mathrm{ClosBa}(X, a)$, and
  (iii) $\overline{1} \in \mathrm{ClosBa}(X, a)$, and
  (iv) $\mathrm{S} \in \mathrm{ClosBa}(X, a)$, and
  (v) $\mathrm{O} \in \mathrm{ClosBa}(X, a)$.

(177) If $\mathrm{I} \in \mathrm{ClosBa}(X, a)$ and $\mathrm{S} \in \mathrm{ClosBa}(X, a)$ and $W \in \mathrm{ClosBa}(X, a)$ and (for every x holds $W(x + 1) = x$) and $F \in \mathrm{ClosBa}(X, a)$, then $F^{\Box} \in \mathrm{ClosBa}(X, a)$.

(178) If $X(0) = 0$ and (for every y such that $0 < y$ holds $X(y) < y$), then $X^{\square(a)}(n) \leq a$.

(179) If $X(0) = 0$ and (for every y such that $0 < y$ holds $X(y) < y$), then $\overline{0} \in \mathrm{ClosBa}(X, a)$.

(180) If $0 < y$, then $\mathrm{E}(y) < y$.

(181) There exists r such that $\mathrm{E}(r \cdot (i+1)) = i$. The theorem is a consequence of (13).

(182) $\mathrm{R}(t+1) = 2 \cdot \mathrm{Q}(\mathrm{R}(t)+4) + \mathrm{R}(t) + 1$.

(183) If (for every x holds $W(x) = x + (a \cdot a + a + 4)$), then $(\mathrm{S}\,(\mathrm{D}\,\mathrm{Q}\,W)^{+})^{\square(a)}(n) + a \cdot a + a = \mathrm{R}(n + a \cdot a)$.

(184) If (for every x holds $W(x) = x + (a \cdot a + a + 4)$) and (for every x holds $X(x) = x + (a+1)$), then $(X\,(\mathrm{S}\,(\mathrm{D}\,\mathrm{Q}\,W)^{+})^{\square(a)})^{\square(a)}(n) = n \cdot n + 2 \cdot a \cdot n + a$.

(185) If (for every n holds $G(n) = n \cdot n + 2 \cdot (c+2) \cdot n + (c+2)$) and (for every x holds $H(x) = x + (c \cdot c + 3 \cdot c + 1)$), then $(\mathrm{E}\,H\,G^{+})(k+1) = k$.

(186) If $2 \leq a$, then $\mathrm{ClosBa}(\mathrm{E}, a) = \mathrm{C}_6$.

(187) $\mathrm{A}(0) = 0$. The theorem is a consequence of (85).

(188) If $0 < y$, then $\mathrm{K}(y) < y$.

(189) There exists r such that $\mathrm{K}(r \cdot (i+1)) = i$. The theorem is a consequence of (85) and (93).

(190) $(\mathrm{S}\,\mathrm{S})(x) = x + 2$.

(191) If (for every x holds $W(x) = x + \mathrm{A}(a)$), then $(\mathrm{S}\,\mathrm{S}\,(\mathrm{K}\,W)^{+})^{\square(a)}(n) + \mathrm{A}(a) + 1 = \mathrm{A}(n + a + 1)$.

(192) If (for every x holds $W(x) = x + \mathrm{A}(a)$) and $G = (\mathrm{S}\,\mathrm{S}\,(\mathrm{K}\,W)^{+})^{\square(a)}$, then $(\mathrm{K}\,W\,G^{+})(k+1) = k$.

(193) If $2 \leq a$, then $\mathrm{ClosBa}(\mathrm{K}, a) = \mathrm{C}_6$.

(194) $\mathrm{L}\,\mathrm{L}^{+} = \overline{0}$.

(195) There exists r such that $\mathrm{L}(r \cdot (i+1)) = i$. The theorem is a consequence of (85).

(196) $\mathrm{L}(0) = 0$. The theorem is a consequence of (187).

(197) If $2 \leq a$, then $\mathrm{ClosBa}(\mathrm{L}, a) = \mathrm{C}_6$.

(198) If $1 \leq a$, then
- (i) $\mathrm{ClosBa}(\mathrm{E}, a) = \mathrm{C}_6$, and
- (ii) $\mathrm{ClosBa}(\mathrm{K}, a) = \mathrm{C}_6$, and
- (iii) $\mathrm{ClosBa}(\mathrm{L}, a) = \mathrm{C}_6$.

13. A Shifted Unary Operator

Let b be a natural number. The functor $\cdot^{+b}$ yielding a unary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 58) for every function F from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x holds $it(F)(x) = F(x) + x + b$.

Let b be a natural number. The functor $\mathrm{ClosVb}(b)$ yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 59) $\mathrm{clos}(\{\mathrm{E}\}, \{\cdot^{+b}, \cdot^{\square}\}, \{\circ\})$.

Now we state the propositions:

(199) $\mathrm{E} \in \mathrm{ClosVb}(b)$. The theorem is a consequence of (5).

(200) If $F \in \mathrm{ClosVb}(b)$, then $F^{+b} \in \mathrm{ClosVb}(b)$. The theorem is a consequence of (6).

(201) If $F \in \mathrm{ClosVb}(b)$, then $F^{\square} \in \mathrm{ClosVb}(b)$. The theorem is a consequence of (6).

(202) If $F \in \mathrm{ClosVb}(b)$ and $G \in \mathrm{ClosVb}(b)$, then $F\,G \in \mathrm{ClosVb}(b)$. The theorem is a consequence of (161).

(203) If $F \in \mathrm{C}_6$, then $F^{+b} \in \mathrm{C}_6$.

(204) $\mathrm{ClosVb}(b) \subseteq \mathrm{C}_6$.

(205) If $1 \leq b$, then $\mathrm{ClosVb}(b) = \mathrm{C}_6$.

(206) $\mathrm{ClosVb}(0) \neq \mathrm{C}_6$.

(207) $\mathrm{ClosVb}(b) = \mathrm{C}_6$ if and only if $1 \leq b$. The theorem is a consequence of (206) and (205).

14. Inversion and General Recursive Functions

From now on e, h, p, z denote natural numbers.

From now on Y denotes a subset of $\mathbb{N}^{\mathbb{N}}$.

Let F be a function from $\mathbb{N}$ into $\mathbb{N}$. We say that F is surj if and only if

(Def. 60) for every natural number x holds there exists a natural number y such that $F(y) = x$.

Let F be a function from $\mathbb{N}$ into $\mathbb{N}$, and x be a natural number. The functor $\mathrm{linv}(F, x)$ yielding a natural number is defined by

(Def. 61) $F(it) = x$ and for every natural number y such that $F(y) = x$ holds $it \leq y$, if F is surj, and $it = F(x)$, otherwise.

The functor $\cdot^{-1}$ yielding a unary operation on $\mathbb{N}^{\mathbb{N}}$ is defined by

(Def. 62) for every function F from $\mathbb{N}$ into $\mathbb{N}$ and every natural number x holds $it(F)(x) = \mathrm{linv}(F, x)$.

Now we state the propositions:

(208) If F is surj, then $F(F^{-1}(x)) = x$.

(209) If F is surj and $F(y) = x$, then $F^{-1}(x) \leq y$.

(210) If F is not surj, then $F^{-1} = F$. The theorem is a consequence of (1).

The functor ClosJR yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 63) $\mathrm{clos}(\{\mathrm{S}, \mathrm{E}\}, \{\cdot^{-1}\}, \{+, \circ\})$.

Let X be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor ClosGR(X) yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 64) $\mathrm{clos}(\{\overline{1}, X\}, \{\cdot^{+}, \cdot^{-1}\}, \{\circ\})$.

Now we state the propositions:

(211) $\mathrm{S} \in \mathrm{ClosJR}$ and $\mathrm{E} \in \mathrm{ClosJR}$. The theorem is a consequence of (5).

(212) If $F \in \mathrm{ClosJR}$, then $F^{-1} \in \mathrm{ClosJR}$. The theorem is a consequence of (6).

(213) If $F \in \mathrm{ClosJR}$ and $G \in \mathrm{ClosJR}$, then $F\,G \in \mathrm{ClosJR}$. The theorem is a consequence of (7).

(214) If $F \in \mathrm{ClosJR}$ and $G \in \mathrm{ClosJR}$, then $F + G \in \mathrm{ClosJR}$. The theorem is a consequence of (7).

(215) $\overline{1} \in \mathrm{ClosGR}(X)$ and $X \in \mathrm{ClosGR}(X)$. The theorem is a consequence of (5).

(216) If $F \in \mathrm{ClosGR}(X)$, then $F^{+} \in \mathrm{ClosGR}(X)$. The theorem is a consequence of (6).

(217) If $F \in \mathrm{ClosGR}(X)$, then $F^{-1} \in \mathrm{ClosGR}(X)$. The theorem is a consequence of (6).

(218) If $F \in \mathrm{ClosGR}(X)$ and $G \in \mathrm{ClosGR}(X)$, then $F\,G \in \mathrm{ClosGR}(X)$. The theorem is a consequence of (161).

(219) If $X \in \mathrm{ClosGR}(W)$, then $\mathrm{ClosGR}(X) \subseteq \mathrm{ClosGR}(W)$.

Let Y be a subset of $\mathbb{N}^{\mathbb{N}}$. We say that Y is rich if and only if

(Def. 65)
(i) $\mathrm{I} \in Y$, and
(ii) $\mathrm{E} \in Y$, and
(iii) $\mathrm{S} \in Y$, and
(iv) $\mathrm{D} \in Y$, and
(v) $\mathrm{M}_3 \in Y$, and
(vi) $\mathrm{Sq} \in Y$, and
(vii) $\mathrm{Hf} \in Y$, and
(viii) $\mathrm{Rt} \in Y$, and
(ix) for every function F from $\mathbb{N}$ into $\mathbb{N}$ such that $F \in Y$ holds $F^{+} \in Y$, and
(x) for every functions F, G from $\mathbb{N}$ into $\mathbb{N}$ such that $F \in Y$ and $G \in Y$ holds $F\,G \in Y$.

Now we state the propositions:

(220) If Y is rich, then
(i) $\mathrm{I} \in Y$, and
(ii) $\mathrm{E} \in Y$, and
(iii) $\mathrm{S} \in Y$, and
(iv) $\mathrm{D} \in Y$, and
(v) $\mathrm{M}_3 \in Y$, and

(vi) $\mathrm{Sq} \in Y$, and
(vii) $\mathrm{Hf} \in Y$, and
(viii) $\mathrm{Rt} \in Y$.

(221) If Y is rich and $F \in Y$, then $F^+ \in Y$.

(222) If Y is rich and $F \in Y$ and $G \in Y$, then $F\,G \in Y$.

(223) If Y is rich and $F \in Y$, then $\mathrm{exprMinusI}(F) \in Y$.

(224) If Y is rich and $F \in Y$ and $G \in Y$, then $\mathrm{exprOplus}(F, G) \in Y$. The theorem is a consequence of (220), (223), (222), and (221).

(225) If Y is rich and $F \in Y$ and $G \in Y$, then $\mathrm{exprOminus}(F, G) \in Y$. The theorem is a consequence of (220), (221), (222), and (223).

(226) If Y is rich, then $\mathrm{exprPhat} \in Y$. The theorem is a consequence of (220), (221), and (225).

(227) If Y is rich and $F \in Y$ and $G \in Y$, then $\mathrm{exprOtimes}(F, G) \in Y$.

(228) If Y is rich and $F \in Y$ and $G \in Y$, then $\mathrm{exprPlus}(F, G) \in Y$.

(229) If Y is rich and $F \in Y$ and $G \in Y$, then $\mathrm{exprMinus}(F, G) \in Y$. The theorem is a consequence of (220), (228), and (222).

(230) If Y is rich and $F \in Y$ and $G \in Y$, then
(i) $F + G \in Y$, and
(ii) if (for every x holds $G(x) \leq F(x)$), then $F \mathbin{\dot{-}} G \in Y$.

(231) $\mathrm{E}(y) \leq 2 \cdot \mathrm{Rt}(y)$. The theorem is a consequence of (12).

(232) $2 \cdot \mathrm{E}(y) \leq y + 1$.

(233) There exists q such that $y = 2 \cdot q$ or $y = 2 \cdot q + 1$.

(234) $(2 \cdot q) \operatorname{div} 2 = q$ and $(2 \cdot q + 1) \operatorname{div} 2 = q$.

(235) E is surj. The theorem is a consequence of (13).

(236) If $x \leq 2 \cdot c$ and $2 \cdot c \leq x + 1$, then $\mathrm{E}^{-1}(x) = c \cdot c + x$.

(237) $\mathrm{E}^{-1}(2 \cdot z) = z \cdot z + 2 \cdot z$. The theorem is a consequence of (236).

(238) $\overline{0}^+ = \mathrm{I}$. The theorem is a consequence of (159).

(239) (i) $\mathrm{S} \in \mathrm{ClosGR(E)}$, and
(ii) $\overline{0} \in \mathrm{ClosGR(E)}$, and
(iii) $\mathrm{I} \in \mathrm{ClosGR(E)}$, and
(iv) $\mathrm{D} \in \mathrm{ClosGR(E)}$, and
(v) $\mathrm{M}_3 \in \mathrm{ClosGR(E)}$.
The theorem is a consequence of (215), (216), (24), (218), (26), (238), (27), and (28).

Let F be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor $\mathrm{exprMinusIE}(F)$ yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 66) $\mathrm{E}\,(\mathrm{S}\,\mathrm{S}\,\mathrm{S}\,\mathrm{M}_3\,F\,\mathrm{E})^+\,\mathrm{S}\,(\mathrm{E}^{-1}\,\mathrm{D}\,F^+)^+$.

Now we state the propositions:

(240) If $F \in \mathrm{ClosGR(E)}$, then $\mathrm{exprMinusIE}(F) \in \mathrm{ClosGR(E)}$.

(241) If $x \leq F(x)$, then $\mathrm{exprMinusIE}(F)(x) = F(x) \mathbin{\dot{-}} x$.

(242) $\mathrm{exprMinusIE}(\mathrm{exprMinusIE}(\mathrm{E}^{-1}\,\mathrm{D})) = \mathrm{Sq}$.

(243) $\mathrm{Sq} \in \mathrm{ClosGR(E)}$. The theorem is a consequence of (215), (217), (239), (218), (240), and (242).

Let F be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor $\mathrm{exprIminusE}(F)$ yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 67) $\mathrm{E}\,(\mathrm{S}\,(F^+)^+\,\mathrm{E})^+\,(\mathrm{Sq}\,F^+)^+$.

Now we state the propositions:

(244) If $F \in \mathrm{ClosGR(E)}$, then $\mathrm{exprIminusE}(F) \in \mathrm{ClosGR(E)}$.

(245) If $F(x) \leq x$, then $\mathrm{exprIminusE}(F)(x) = x \dot{-} F(x)$.

The functor exprW yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 68) $\mathrm{E}\,(\mathrm{S\,S\,D\,E\,E})^+\,\mathrm{E}^{-1}$.

Now we state the propositions:

(246) $\mathrm{exprW} \in \mathrm{ClosGR(E)}$.

(247) If $\mathrm{E}(c \cdot c + y) = y$, then $(\mathrm{S\,S\,D\,E\,E})^+(c \cdot c + y) = 2 \cdot \mathrm{E}(y) + 2 + (c \cdot c + y)$.

(248) $\mathrm{exprW}(2 \cdot q) = 2 \cdot \mathrm{E}(2 \cdot q) + 1$ and $\mathrm{exprW}(2 \cdot q + 1) = 2 \cdot \mathrm{E}(2 \cdot q + 1)$.

(249) exprW is surj.

(250) $\mathrm{E}\,\mathrm{exprW}^{-1} = \mathrm{Hf}$.

(251) $\mathrm{Hf} \in \mathrm{ClosGR(E)}$. The theorem is a consequence of (215), (246), (217), (218), and (250).

(252) $\mathrm{Rt}(4 \cdot x + 1) = 2 \cdot \mathrm{Rt}(x)$ or $\mathrm{Rt}(4 \cdot x + 1) = 2 \cdot \mathrm{Rt}(x) + 1$.

The functor exprRtGE yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 69) $\mathrm{Hf\,Hf\,S\,S\,E\,exprIminusE}(\overline{1})\,\mathrm{exprIminusE(E)\,S\,M}_3^+$.

Now we state the propositions:

(253) $\mathrm{exprRtGE} = \mathrm{Rt}$.

(254) $\mathrm{Rt} \in \mathrm{ClosGR(E)}$.

(255) $\mathrm{ClosGR(E)}$ is rich. The theorem is a consequence of (239), (215), (243), (251), (254), (216), and (218).

(256) If Y is rich, then
- (i) $\mathrm{A} \in Y$, and
- (ii) $\mathrm{V} \in Y$, and
- (iii) $\mathrm{K} \in Y$, and
- (iv) $\mathrm{L} \in Y$.

(257)
- (i) $\mathrm{I} \in \mathrm{ClosJR}$, and
- (ii) $\overline{0} \in \mathrm{ClosJR}$, and
- (iii) $\overline{1} \in \mathrm{ClosJR}$.

(258) $\mathrm{ClosGR(E)} \subseteq \mathrm{ClosJR}$.

(259) If Y is rich and (for every F such that $F \in Y$ holds $F^{-1} \in Y$), then $\mathrm{ClosJR} \subseteq Y$.

(260) $\mathrm{ClosGR(E)} = \mathrm{ClosJR}$. The theorem is a consequence of (217), (255), (259), and (258).

(261) (i) $\mathrm{S} \in \mathrm{ClosGR}(\mathrm{K})$, and
(ii) $\overline{0} \in \mathrm{ClosGR}(\mathrm{K})$, and
(iii) $\mathrm{I} \in \mathrm{ClosGR}(\mathrm{K})$, and
(iv) $\mathrm{D} \in \mathrm{ClosGR}(\mathrm{K})$, and
(v) $\mathrm{M}_3 \in \mathrm{ClosGR}(\mathrm{K})$.
The theorem is a consequence of (215), (216), (24), (218), (111), (238), (27), and (28).

(262) K is surj. The theorem is a consequence of (93).

(263) $\mathrm{K}(\mathrm{Z}(x)) = x$. The theorem is a consequence of (93).

(264) $\mathrm{K}^{-1} = \mathrm{Z}$.

(265) $\mathrm{Z} \in \mathrm{ClosGR}(\mathrm{K})$. The theorem is a consequence of (215), (217), and (264).

The functor exprAK yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 70) $\mathrm{K}\,\mathrm{S}\,\mathrm{S}\,\mathrm{S}\,((\mathrm{Z}\,\mathrm{Z}^+\,\mathrm{K})^+\,\mathrm{Z}\,\mathrm{K})^+\,\mathrm{Z}$.

Now we state the propositions:

(266) $\mathrm{exprAK} = \mathrm{A}$.

(267) $\mathrm{A} \in \mathrm{ClosGR}(\mathrm{K})$.

Let F be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor exprMinusIK(F) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 71) $\mathrm{K}\,(\mathrm{S}\,\mathrm{S}\,\mathrm{S}\,\mathrm{M}_3\,F\,\mathrm{K})^+\,(\mathrm{A}\,F^+)^+$.

Now we state the propositions:

(268) If $F \in \mathrm{ClosGR}(\mathrm{K})$, then $\mathrm{exprMinusIK}(F) \in \mathrm{ClosGR}(\mathrm{K})$.

(269) If $x \leq F(x)$, then $\mathrm{exprMinusIK}(F)(x) = F(x) \dot{-} x$.

(270) $\mathrm{exprMinusIK}(\mathrm{exprMinusIK}(\mathrm{exprMinusIK}(\mathrm{D}\,\mathrm{Z}))) = \mathrm{Sq}$.

(271) $\mathrm{Sq} \in \mathrm{ClosGR}(\mathrm{K})$. The theorem is a consequence of (261), (265), (218), (268), and (270).

The functor exprPK yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 72) $\mathrm{K}\,(\mathrm{K}\,\mathrm{S})^{-1}$.

Now we state the propositions:

(272) $\mathrm{K}\,\mathrm{S}$ is surj. The theorem is a consequence of (86) and (93).

(273) $\mathrm{exprPK}(0) = 0$ and $\mathrm{exprPK}(x+1) = x$.

(274) $\mathrm{exprPK} \in \mathrm{ClosGR}(\mathrm{K})$. The theorem is a consequence of (215), (261), (218), and (217).

(275) If $F \in \mathrm{ClosGR}(\mathrm{K})$, then $\mathrm{exprIminusF}(F) \in \mathrm{ClosGR}(\mathrm{K})$.

(276) $\mathrm{exprIminusF}(\mathrm{K})(y) = \mathrm{A}(\mathrm{V}(y))$. The theorem is a consequence of (94).

The functor exprU yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 73) $\mathrm{exprIminusF}(\mathrm{A}\,\mathrm{K}\,\mathrm{exprPK}\,\mathrm{exprIminusF}(\mathrm{K}))$.

Now we state the propositions:

(277) $\mathrm{exprU} \in \mathrm{ClosGR}(\mathrm{K})$. The theorem is a consequence of (267), (215), (274), (218), and (275).

(278) $\mathrm{exprU}(y) = \mathrm{V}(y) + \mathrm{K}(y)$.

(279) If $k \leq t$, then $\operatorname{exprU}(\mathrm{A}(t) + k) = t + k$. The theorem is a consequence of (93) and (278).

(280) exprU is surj.

(281) $\mathrm{K} \operatorname{exprU}^{-1} = \mathrm{Hf}$.

(282) $\mathrm{Hf} \in \operatorname{ClosGR}(\mathrm{K})$. The theorem is a consequence of (215), (277), (217), (218), and (281).

The functor exprB yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 74) $\operatorname{exprIminusF}(\operatorname{exprIminusF}(\operatorname{Sq} \operatorname{exprIminusF}(\mathrm{Hf}) \operatorname{exprU}) \operatorname{exprIminusF}(\mathrm{K}))$.

Now we state the propositions:

(283) $\mathrm{exprB} \in \operatorname{ClosGR}(\mathrm{K})$.

(284) If ($r = 0$ or $r = 1$), then $\mathrm{A}(2 \cdot h + r) = (h + r) \cdot (h + r) + (h \cdot h + h)$.

(285) If ($r = 0$ or $r = 1$) and $\mathrm{V}(y) = 2 \cdot h + r$, then $\operatorname{exprB}(y) = \mathrm{K}(y) + (h + r) \cdot (h + r)$.

(286) exprB is surj. The theorem is a consequence of (231) and (93).

(287) $\mathrm{K} \operatorname{exprB}^{-1} = \mathrm{E}$.

(288) $\mathrm{E} \in \operatorname{ClosGR}(\mathrm{K})$. The theorem is a consequence of (215), (283), (217), (218), and (287).

(289) (i) $\operatorname{ClosGR}(\mathrm{E}) = \operatorname{ClosJR}$, and
(ii) $\operatorname{ClosGR}(\mathrm{K}) = \operatorname{ClosJR}$, and
(iii) $\operatorname{ClosGR}(\mathrm{L}) = \operatorname{ClosJR}$.

15. THE CONSTANT $\overline{1}$ AND THE SUCCESSOR

Now we state the propositions:

(290) If $I \subseteq \mathbb{N}^{\mathbb{N}}$ and (for every F such that $F \in I$ holds $F(0) = 0$) and (for every unary operation u on $\mathbb{N}^{\mathbb{N}}$ and F such that $u \in U$ and $F(0) = 0$ holds $u(F)(0) = 0$) and $F \in \operatorname{clos}(I, U, \{\circ\})$, then $F(0) = 0$.

(291) If $F(0) = 0$, then $F^{-1}(0) = 0$. The theorem is a consequence of (209) and (210).

(292) If $F(0) = 0$, then $F^{+}(0) = 0$.

(293) $F^{\square}(0) = 0$.

(294) $\{X\} \subseteq \mathbb{N}^{\mathbb{N}}$ and $\{W, X\} \subseteq \mathbb{N}^{\mathbb{N}}$.

(295) (i) $\mathrm{E}(0) = 0$, and
(ii) $\mathrm{K}(0) = 0$, and
(iii) $\mathrm{L}(0) = 0$.
The theorem is a consequence of (22), (149), and (196).

Let X be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor $\operatorname{ClosGR0}(X)$ yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 75) $\operatorname{clos}(\{X\}, \{\cdot^{+}, \cdot^{-1}\}, \{\circ\})$.

Now we state the propositions:

(296) If $X(0) = 0$ and $F \in \operatorname{ClosGR0}(X)$, then $F(0) = 0$.

(297) (i) ClosGR0(E) $\neq$ ClosJR, and
(ii) ClosGR0(K) $\neq$ ClosJR, and
(iii) ClosGR0(L) $\neq$ ClosJR.

The theorem is a consequence of (257), (295), and (296).

Let X be a function from $\mathbb{N}$ into $\mathbb{N}$. The functor ClosGRS(X) yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 76) clos($\{\mathrm{S}, X\}, \{\cdot^+, \cdot^{-1}\}, \{\circ\}$).

Now we state the propositions:

(298) (i) $\mathrm{S} \in \mathrm{ClosGRS}(X)$, and
(ii) $X \in \mathrm{ClosGRS}(X)$, and
(iii) if $F \in \mathrm{ClosGRS}(X)$, then $F^+ \in \mathrm{ClosGRS}(X)$ and $F^{-1} \in \mathrm{ClosGRS}(X)$, and
(iv) if $F \in \mathrm{ClosGRS}(X)$ and $G \in \mathrm{ClosGRS}(X)$, then $F\,G \in \mathrm{ClosGRS}(X)$.

The theorem is a consequence of (5), (6), and (161).

(299) If $\overline{0} \in \mathrm{ClosGRS}(X)$, then $\mathrm{ClosGRS}(X) = \mathrm{ClosGR}(X)$.

(300) $(\mathrm{E}\,\mathrm{S}\,\mathrm{E}^{-1})(2 \cdot q) = 0$ and $(\mathrm{E}\,\mathrm{S}\,\mathrm{E}^{-1})(2 \cdot q + 1) = 2 \cdot (q + 1)$.

(301) L S is surj. The theorem is a consequence of (93).

(302) $(\mathrm{L}\,\mathrm{S})^{-1}(x + 1) = \mathrm{A}(x) + x$.

(303) (i) $\overline{0} \in \mathrm{ClosGRS}(\mathrm{E})$, and
(ii) $\overline{0} \in \mathrm{ClosGRS}(\mathrm{K})$, and
(iii) $\overline{0} \in \mathrm{ClosGRS}(\mathrm{L})$.

(304) (i) ClosGRS(E) = ClosJR, and
(ii) ClosGRS(K) = ClosJR, and
(iii) ClosGRS(L) = ClosJR.

The theorem is a consequence of (303), (299), and (289).

16. The Shifted Operator with Inversion

From now on g, v, n_1, c_1 denote natural numbers.

Let X be a function from $\mathbb{N}$ into $\mathbb{N}$, and b be a natural number. The functor ClosGRb(X, b) yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 77) clos($\{X\}, \{\cdot^{+b}, \cdot^{-1}\}, \{\circ\}$).

Now we state the propositions:

(305) (i) $X \in \mathrm{ClosGRb}(X, b)$, and
(ii) if $F \in \mathrm{ClosGRb}(X, b)$, then $F^{+b} \in \mathrm{ClosGRb}(X, b)$ and $F^{-1} \in \mathrm{ClosGRb}(X, b)$, and
(iii) if $F \in \mathrm{ClosGRb}(X, b)$ and $G \in \mathrm{ClosGRb}(X, b)$, then $F\,G \in \mathrm{ClosGRb}(X, b)$.

(306) If $W \in \mathrm{ClosGRb}(X, b)$ and (for every x holds $W(x + 1) = x$) and $F \in \mathrm{ClosGRb}(X, b)$, then $F^+ \in \mathrm{ClosGRb}(X, b)$.

(307) If $\overline{1} \in \mathrm{ClosGRb}(X, b)$ and $W \in \mathrm{ClosGRb}(X, b)$ and (for every x holds $W(x + 1) = x$), then $\mathrm{ClosGR}(X) \subseteq \mathrm{ClosGRb}(X, b)$.

(308) If $F \in \mathrm{ClosGR(E)}$, then $F^{+b} \in \mathrm{ClosGR(E)}$.

(309) If $X \in \mathrm{ClosGR(E)}$, then $\mathrm{ClosGRb}(X, b) \subseteq \mathrm{ClosGR(E)}$.

(310) $\mathrm{I}^{+b}(x) = 2 \cdot x + b$.

(311) $\mathrm{E}\,\mathrm{E}^{-1} = \mathrm{I}$. The theorem is a consequence of (1).

(312) $\mathrm{E}^{-1}(2 \cdot m + 1) = (m+1) \cdot (m+1) + (2 \cdot m + 1)$. The theorem is a consequence of (236).

(313) If (for every y holds $W(y) = y + c$) and $c \leq x$, then $\mathrm{E}\,W$ is surj and $\mathrm{E}((\mathrm{E}\,W)^{-1}(x)) = x \dot{-} c$.

(314) (i) if $x + c \leq 2 \cdot (m+1)$, then $\mathrm{E}(\mathrm{E}^{-1}(2 \cdot m) + x + (c+1)) = x + c$, and
(ii) if $x + c \leq 2 \cdot (m+2)$, then $\mathrm{E}(\mathrm{E}^{-1}(2 \cdot m + 1) + x + (c+2)) = x + c$.

Let b be a natural number. The functor $\mathrm{exprW1}(b)$ yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 78) $\mathrm{E}\,(\mathrm{E}^{-1}\,\mathrm{I}^{+b})^{+b}$.

Let b be a natural number. The functor $\mathrm{exprW2}(b)$ yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 79) $\mathrm{E}\,(\mathrm{E}^{-1}\,\mathrm{exprW1}(b)\,\mathrm{I}^{+b})^{+b}$.

Now we state the propositions:

(315) $\mathrm{exprW1}(b)(x) = \mathrm{E}(\mathrm{E}^{-1}(2 \cdot x + b) + x + b)$ and $\mathrm{exprW2}(b)(x) = \mathrm{E}(\mathrm{E}^{-1}(\mathrm{exprW1}(b)(2 \cdot x + b)) + x + b)$.

(316) (i) if $b = 2 \cdot p + 2$, then $\mathrm{exprW1}(b)(x) = x + (2 \cdot p + 1)$ and $\mathrm{exprW2}(b)(x) = x + 2 \cdot p$, and
(ii) if $b = 2 \cdot p + 3$, then $\mathrm{exprW1}(b)(x) = x + (2 \cdot p + 1)$ and $\mathrm{exprW2}(b)(x) = x + (2 \cdot p + 2)$.

(317) (i) $\mathrm{I} \in \mathrm{ClosGRb(E}, b)$, and
(ii) $\mathrm{exprW1}(b) \in \mathrm{ClosGRb(E}, b)$, and
(iii) $\mathrm{exprW2}(b) \in \mathrm{ClosGRb(E}, b)$.

(318) If $b = d + 2$, then there exist G, H such that $G \in \mathrm{ClosGRb(E}, b)$ and $H \in \mathrm{ClosGRb(E}, b)$ and for every x holds $G(x) = x + (d+1)$ and $H(x) = x + d$.

The functor exprPE1 yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 80) $\mathrm{exprW1}(1)$.

The functor exprSE1 yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 81) $\mathrm{E}\,\mathrm{exprPE1}^{-1}\,(\mathrm{E}^{-1}\,\mathrm{exprPE1}\,\mathrm{I}^{+1})^{+1}$.

Now we state the propositions:

(319) $\mathrm{exprPE1}(0) = 2$ and $\mathrm{exprPE1}(y+1) = y$.

(320) If $v \neq 2$, then $\mathrm{exprPE1}^{-1}(v) = v + 1$.

(321) $\mathrm{exprSE1} = \mathrm{S}$.

(322) If $1 \leq b$, then $\mathrm{S} \in \mathrm{ClosGRb(E}, b)$.

(323) $\mathrm{E}\,\mathrm{S}\,\mathrm{E}^{-1}\,\mathrm{I}^{+} = \overline{0}$.

(324) If $1 \leq b$, then $\mathrm{ClosGRb(E}, b) = \mathrm{ClosJR}$.

(325) $n \leq \mathrm{A}(n)$. The theorem is a consequence of (57) and (85).

(326) If (for every y holds $W(y) = y + c$) and $c \leq x$, then $\mathrm{K}\,W$ is surj and $\mathrm{K}((\mathrm{K}\,W)^{-1}(x)) = x \dot{-} c$.

Let b be a natural number. The functor exprUK(b) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 82) $\mathrm{K}\,(\mathrm{Z}\,\mathrm{I}^{+b})^{+b}$.

Now we state the propositions:

(327) If $b = c + 1$, then $\mathrm{exprUK}(b)(x) = x + c$.

(328) (i) $\mathrm{Z} \in \mathrm{ClosGRb}(\mathrm{K}, b)$, and
(ii) $\mathrm{I} \in \mathrm{ClosGRb}(\mathrm{K}, b)$, and
(iii) $\mathrm{exprUK}(b) \in \mathrm{ClosGRb}(\mathrm{K}, b)$.

Let b be a natural number. The functor exprP1K(b) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 83) $\mathrm{K}\,(\mathrm{K}\,\mathrm{exprUK}(b))^{-1}$.

Let b be a natural number. The functor exprCK(b) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 84) $\mathrm{K}\,(\mathrm{Z}\,\mathrm{exprP1K}(b))^{+b}\,\mathrm{exprUK}(b)\,\mathrm{exprUK}(b)\,\mathrm{exprUK}(b)$.

Now we state the propositions:

(329) If $b = c + 1$, then
(i) $(\mathrm{exprUK}(b)\,\mathrm{exprUK}(b)\,\mathrm{exprUK}(b))(x) = x + 3 \cdot c$, and
(ii) for every G holds $(G\,\mathrm{exprUK}(b)\,\mathrm{exprUK}(b)\,\mathrm{exprUK}(b))(x) = G(x+3\cdot c)$.

(330) If $b = d + 2$, then $\mathrm{exprCK}(b)(x) = 2 \cdot d$.

Let b be a natural number. The functor exprSK2(b) yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 85) $\mathrm{K}\,(\mathrm{K}\,\mathrm{exprCK}(b)^{+b})^{-1}\,\mathrm{exprUK}(b)\,\mathrm{exprUK}(b)\,\mathrm{exprUK}(b)$.

Now we state the proposition:

(331) If $b = d + 2$, then $\mathrm{exprSK2}(b) = \mathrm{S}$.

The functor exprPK1 yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 86) $\mathrm{K}\,(\mathrm{Z}^{+1})^{+1}$.

The functor exprSK1 yielding a function from $\mathbb{N}$ into $\mathbb{N}$ is defined by the term

(Def. 87) $\mathrm{K}\,\mathrm{exprPK1}^{-1}\,(\mathrm{Z}\,\mathrm{I}^{+1})^{+1}$.

Now we state the propositions:

(332) $\mathrm{exprPK1}(0) = 1$ and $\mathrm{exprPK1}(y + 1) = y$.

(333) If $v \neq 1$, then $\mathrm{exprPK1}^{-1}(v) = v + 1$.

(334) $\mathrm{exprSK1} = \mathrm{S}$.

(335) If $1 \leq b$, then $\mathrm{S} \in \mathrm{ClosGRb}(\mathrm{K}, b)$.

(336) If $1 \leq b$, then $\mathrm{ClosGRb}(\mathrm{K}, b) = \mathrm{ClosJR}$.

(337) L is surj and $\mathrm{L}^{-1} = \mathrm{A}$.

(338) $\mathrm{A}(x + n) = \mathrm{A}(x) + n \cdot x + \mathrm{A}(n)$. The theorem is a consequence of (85).

(339) There exists H such that $H \in \mathrm{ClosGRb}(\mathrm{L}, b)$ and for every x holds $H(x) = \mathrm{A}(x) + n \cdot (x + b)$.

(340) If $1 \leq b$, then $\mathrm{S} \in \mathrm{ClosGRb}(\mathrm{L}, b)$.

(341) If $1 \leq b$, then $\mathrm{ClosGRb}(\mathrm{L}, b) = \mathrm{ClosJR}$.

(342) If $X(0) = 0$ and $F \in \mathrm{ClosGRb}(X, 0)$, then $F(0) = 0$.

(343) (i) $\mathrm{ClosGRb}(\mathrm{E}, b) = \mathrm{ClosJR}$ if and only if $1 \leq b$, and
(ii) $\mathrm{ClosGRb}(\mathrm{K}, b) = \mathrm{ClosJR}$ if and only if $1 \leq b$, and
(iii) $\mathrm{ClosGRb}(\mathrm{L}, b) = \mathrm{ClosJR}$ if and only if $1 \leq b$.

The theorem is a consequence of (257), (295), (342), (324), (336), and (341).

17. Any Constant $\overline{n}$, $n \geq 1$

Let X be a function from $\mathbb{N}$ into $\mathbb{N}$, U be a set, and n be a natural number. The functor $\mathrm{ClosC}(X, U, n)$ yielding a subset of $\mathbb{N}^{\mathbb{N}}$ is defined by the term

(Def. 88) $\mathrm{clos}(\{\overline{n}, X\}, U, \{\circ\})$.

Now we state the propositions:

(344) (i) $\overline{n} \in \mathrm{ClosC}(X, U, n)$, and
(ii) $X \in \mathrm{ClosC}(X, U, n)$, and
(iii) if $\cdot^+ \in U$ and $F \in \mathrm{ClosC}(X, U, n)$, then $F^+ \in \mathrm{ClosC}(X, U, n)$, and
(iv) if $F \in \mathrm{ClosC}(X, U, n)$ and $G \in \mathrm{ClosC}(X, U, n)$, then $F\,G \in \mathrm{ClosC}(X, U, n)$.

The theorem is a consequence of (5), (6), and (161).

(345) $F\,\overline{m} = \overline{F(m)}$. The theorem is a consequence of (1).

(346) If $\cdot^+ \in U$ and $\overline{c} \in \mathrm{ClosC}(X, U, n)$, then $\overline{c + k \cdot n} \in \mathrm{ClosC}(X, U, n)$ and $\overline{X(c + k \cdot n)} \in \mathrm{ClosC}(X, U, n)$.

(347) If $1 \leq n$ and (for every c such that $\overline{c} \in \mathrm{ClosC}(X, U, n)$ and $1 \leq c$ holds $\overline{c \dot{-} 1} \in \mathrm{ClosC}(X, U, n)$), then $\overline{1} \in \mathrm{ClosC}(X, U, n)$.

(348) If $\cdot^+ \in U$ and $1 \leq n$ and $\overline{1} \in \mathrm{ClosC}(X, U, n)$, then $\mathrm{ClosC}(X, U, n) = \mathrm{ClosC}(X, U, 1)$.

(349) If $n = n_1 + 1$ and $c = c_1 + 1$, then $\mathrm{E}(c + (n \cdot c \cdot c + 2 \cdot c) \cdot n) = c_1$. The theorem is a consequence of (13).

(350) If $n = n_1 + 1$ and $c = c_1 + 1$, then $\mathrm{K}(c + (2 \cdot n \cdot c \cdot c + 3 \cdot c) \cdot n) = c_1$. The theorem is a consequence of (85) and (93).

(351) If $n = n_1 + 1$, then $\mathrm{L}(c + (2 \cdot n \cdot c \cdot c + 2 \cdot c) \cdot n) = n_1 \cdot c$. The theorem is a consequence of (85).

(352) If $n = n_1 + 1$ and $c = c_1 + 1$, then $\mathrm{L}(n_1 \cdot c + (2 \cdot n \cdot c \cdot c) \cdot n) = c_1$.

(353) If $\cdot^+ \in U$ and $1 \leq n$, then
(i) $\overline{1} \in \mathrm{ClosC}(\mathrm{E}, U, n)$, and
(ii) $\overline{1} \in \mathrm{ClosC}(\mathrm{K}, U, n)$, and
(iii) $\overline{1} \in \mathrm{ClosC}(\mathrm{L}, U, n)$.

(354) If $\cdot^+ \in U$ and $1 \leq n$, then
(i) $\mathrm{ClosC}(\mathrm{E}, U, n) = \mathrm{ClosC}(\mathrm{E}, U, 1)$, and
(ii) $\mathrm{ClosC}(\mathrm{K}, U, n) = \mathrm{ClosC}(\mathrm{K}, U, 1)$, and
(iii) $\mathrm{ClosC}(\mathrm{L}, U, n) = \mathrm{ClosC}(\mathrm{L}, U, 1)$.

The theorem is a consequence of (353) and (348).

(355) If $X(0) = 0$ and (for every unary operation u on $\mathbb{N}^{\mathbb{N}}$ and G such that $u \in U$ and $G(0) = 0$ holds $u(G)(0) = 0$) and $F \in \mathrm{ClosC}(X, U, 0)$, then $F(0) = 0$.

(356) (i) for every unary operation u on $\mathbb{N}^{\mathbb{N}}$ and G such that $u \in \{\cdot^{+}, \cdot^{\square}\}$ and $G(0) = 0$ holds $u(G)(0) = 0$, and
(ii) for every unary operation u on $\mathbb{N}^{\mathbb{N}}$ and G such that $u \in \{\cdot^{+}, \cdot^{-1}\}$ and $G(0) = 0$ holds $u(G)(0) = 0$.

(357) (i) $\mathrm{ClosC}(\mathrm{E}, \{\cdot^{+}, \cdot^{\square}\}, n) = \mathrm{C}_6$ if and only if $1 \leq n$, and
(ii) $\mathrm{ClosC}(\mathrm{K}, \{\cdot^{+}, \cdot^{\square}\}, n) = \mathrm{C}_6$ if and only if $1 \leq n$, and
(iii) $\mathrm{ClosC}(\mathrm{L}, \{\cdot^{+}, \cdot^{\square}\}, n) = \mathrm{C}_6$ if and only if $1 \leq n$.

(358) (i) $\mathrm{ClosC}(\mathrm{E}, \{\cdot^{+}, \cdot^{-1}\}, n) = \mathrm{ClosJR}$ if and only if $1 \leq n$, and
(ii) $\mathrm{ClosC}(\mathrm{K}, \{\cdot^{+}, \cdot^{-1}\}, n) = \mathrm{ClosJR}$ if and only if $1 \leq n$, and
(iii) $\mathrm{ClosC}(\mathrm{L}, \{\cdot^{+}, \cdot^{-1}\}, n) = \mathrm{ClosJR}$ if and only if $1 \leq n$.

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